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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: MATERIALS AND METHODS Animal experiments Animal experiments performed in this study were approved by the Chinese Academy of Sciences Institutional Animal Care and Use Committee.
IRB: In brief , primary hMSCs were isolated under the approval of the ethics committee of 306th Hospital of PLA in Beijing .
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Then, the mixture was injected subcutaneously into the inguinal region of NODSCID mice (male, 6 weeks old).
Subject Demographics
Age: Then, the mixture was injected subcutaneously into the inguinal region of NODSCID mice (male, 6 weeks old).
Randomization
To eliminate bias introduced by sequencing depth , replicates for each sample were merged and the same numbers of high-quality reads were randomly sampled for merged bams .
Blinding
not detected.
Power Analysis
not detected.
Replication
Each point of the standard curve and the cell samples were measured in triplicate.

Number: Each point of the standard curve and the cell samples were measured in triplicate.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Antibodies used for immunofluorescence analysis are as follows: anti-SOX2 (Santa Cruz , sc-17320) , anti-OCT4 (Santa Cruz , sc-5279) , anti-NANOG (Abcam , ab21624) , anti-Ki67 (ZSGB-BIO , ZM-0166) , anti-gH2AX (Millipore , 05-636) , anti-53BP1 (BETYHL , A300-273A) , anti-Lamin	anti-SOX2		
	anti-OCT4		
	anti-NANOG		
	anti-Ki67		
	anti-gH2AX		
	anti-53BP1		
	anti-Lamin		
Antibodies used for western blot analysis are as follows: anti-SIRT3 (Cell Signaling Technology , 2627) , anti-SIRT3 (Cell Signaling Technology , 5490) , anti-SIRT3 (Abcam , ab189860) , anti-SIRT3 (Sigma , S4072) , anti-SIRT3 (Abgent , AP6242a) , anti-P16 (BD Bioscience , 550834) , anti-P21 (Cell Signaling Technology , 2947) , anti-b-Tubulin (Immunoway , YM3030) ,	anti-SIRT3 (Cell Signaling Technology , 2627		
	anti-SIRT3 (Cell Signaling Technology , 5490		
	anti-SIRT3		
	S4072		
	anti-P16		
	anti-P21 (Cell Signaling Technology , 2947)		
	anti-b-Tubulin		
Primary antibodies used were anti-FLAG (Sigma , F1804) and anti-Lamin B1 (Abcam , ab16048) .	anti-FLAG		
	anti-Lamin B1		
Experimental Models: Cell Lines			
In brief , HEK293T cells transfected with plasmids expressing SIRT3-FLAG or Luc-FLAG were lysed in CHAPS lysis buffer (120 mM NaCl , 0.3 CHAPS (Sigma , C900480) , 1 mM EDTA , 40 mM HEPES , pH 7.5 , and cOmpleteTM protease inhibitor cocktail (Roche)) at 4degC for 2 h with rotation , and then centrifuged at 12 000 g at 4degC for 30 min to collect the supernatants .	HEK293T		Suggestion: RRID:CVCL_0063(link)
Experimental Models: Organisms/Strains			
Then, the mixture was injected subcutaneously into the inguinal	NODSCID		Suggestion: RRID:IMSR_ARC:NODSCID(link)

region of NODSCID mice (male, 6 weeks old).			
Recombinant DNA			
Sequences representing sgRNAs targeting exon 1 or exon 3 of SIRT3 were cloned into pCAG-mCherry-sgRNA vector (Addgene, 87110).	pCAG-mCherry-sgRNA		
After treatment with ROCK inhibitor (Y-27632, TOCRIS) for 24 h, pCAG-mCherry-SIRT3 sgRNAs and pCAG-1BPNLS-Cas9-1BPNLS-2AGFP were simultaneously electroporated into SIRT3++ hESCs by 4D-Nucleofector (Lonza).	pCAG-mCherry-SIRT3		
	pCAG-1BPNLS-Cas9-1BPNLS-2AGFP		Suggestion: RRID:Addgene_87109(link)
Genomic DNA fragment of the OCT4 promoter region was amplified using LA Taq DNA Polymerase Hot-Start Version (TaKaRa) , and subsequently cloned into the pMD20 T vector (TaKaRa) .	pMD20 T		
Lentivirus production HEK293T cells were co-transfected with indicated lentiviral overexpression plasmids and lentiviral packing vectors including psPAX2 (Addgene , 12260) and pMD2 . G (Addgene , 12259) .	psPAX2		
	pMD2 . G		
Plasmid construction SIRT3-FLAG cDNA was generated from the cDNA of SIRT3++ hMSCs via PCR amplification and then cloned into the pLE4 empty vector (a kind gift from Dr. Tomoaki Hishida) by using NovoRec(r) plus One-step PCR Cloning Kit (Novoprotein , NR005) according to the manufacturers instructions and confirmed by DNA sequencing .	pLE4		
Software and Algorithms			
Mitochondrial number per cell and individual mitochondrial size were analyzed from 30 to 40 electron micrographs using ImageJ-based Fiji software (https:imagej . netFiji) (42) .	Fiji		
The relative cell density was quantified using ImageJ .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
Raw data were processed with the MaxQuant (Version 1.3) and Proteome Discoverer (Version 1.4) .	Proteome Discoverer		Suggestion: Proteome Discoverer, RRID:SCR_014477(link)
False discovery rate (FDR) was analyzed with Percolator and set	Percolator		

to 1 as the threshold of the peptide confidence parameter .			
Gene Ontology Cellular Component (GO-CC) terms analysis was performed using Metascape (45) .	Metascape		
Using Imaris software , the Surfaces tool was used to create a 3D rendering of each channel .	Imaris		
First , low-quality reads and adaptors were removed using TrimGalore (v0.4.4_dev) (54) .	TrimGalore		
Clean reads were aligned to the UCSC human hg19 genome using HISAT2 software (v2.1.0) (54) .	HISAT2		
Counts for annotated genes were generated using HTSeq (v0.6.1) (55) .	HTSeq		
The RBioconductor package DESeq2 (56) was used to identify differentially expressed genes with a cutoff Benjamini-Hochberg adjust P-value less than 0.05 and log2 (fold change) > 0.58 .	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
Enrichment analysis was performed using Toppgene (57) (https:toppgene.cchmc . org) with adjusted P-value < 0.05 .	Toppgene		
Unique and multi-mapped reads were assigned to repetitive elements using RepEnrich2	RepEnrich2		
Differentially expressed repetitive elements were analyzed with edgeR (58) on the fractional counts data .	edgeR		
Duplicate reads were also removed using Picard for future analysis .	Picard		
Briefly , the annotations of repetitive elements (from HOMER) or protein-coding genes were intersected with LAD regions using the findOverlapPairs function in RBioconductor package GenomicRanges (v1.36.0)	RBioconductor		
	GenomicRanges		
Determination of expression levels of LAD-localized repetitive elements To determine the expression levels of repetitive elements within LAD regions , cleaned reads from RNA-seq were realigned to the human hg19 genome using STAR (v5.21) as previously described (61) .	STAR		
Clean reads were then aligned to the human hg19 genome using Bowtie2 (v2.2.9) with default parameters .	Bowtie2		
Reads aligned to mitochondrial DNA (chrM) or with low mapping quality	samtools		

(MAPQ score < 10) were removed using samtools (v1.9) (62) .			
Each read was extended by 250 bp and normalized by RPKM for each 10-bp using bamCoverage function in deepTools (v3.3.0) software (63) .	deepTools		
Genome annotations for ATAC-seq peaks were performed using the annotatePeaks function in HOMER software (65).	HOMER		
The RBioconductor package DiffBind (v2.12.0) (66) was used to identify differential peaks occupancy with the DESeq2 method.	DiffBind		
Two-tailed Students t-test was conducted using the GraphPad Prism 8 software.	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
The RBioconductor package DESeq2 (56) was used to identify differentially expressed genes with a cutoff Benjamini-Hochberg adjust P-value less than 0.05 and log2 (fold change) > 0.58 .	Benjamini-Hochberg
Two-tailed Students t-test was conducted using the GraphPad Prism 8 software.	Students t-test

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